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QLSTM4FM: Quantum Assisted Long Short-Term Memory Framework for Financial Market Trend Forecasting

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***Abstract*—This research explores a novel hybrid neural network architecture for stock price forecasting that integrates Variational Quantum Layers (VQLs) to enhance efficiency and trainability. It examines various quantum-classical algorithms through extensive simulations on both classical and quantum hardware, utilizing key stock indicators like MACD, EMA, Bollinger Bands, and momentum metrics, alongside sentiment analysis of financial news. The dataset used includes real-time stock prices for 15 companies, focusing on Merck & Co. (MRK). By framing stock price forecasting as a binary classification problem, It proposes *QLSTM4FM* i.e. a unique framework based on quantum-assisted LSTM model, which significantly enhances stock price forecasting. It conducts a thorough and comprehensive performance evaluation of the proposed framework against classical LSTM models. Results show that *QLSTM4FM*, featuring quantum-enhanced layers, outperforms classical LSTM models in accuracy, lower loss, and faster convergence, demonstrating the potential of quantum computing in financial forecasting.**

***Index Terms*—Quantum Finance, Variational Quantum Layers (VQLs), QLSTM, Sentiment Analysis**

I. Introduction

Since the 1980s, quantum computing has intrigued academicians for its potential to solve complex computational problems. One of the initial individuals to propose that quantum physics may perform better than conventional computation was Richard Feynman, leading to its integration into neural networks [7]. While classical machine learning, particularly deep neural networks (DNNs), has excelled at detecting data patterns, quantum mechanics is uniquely positioned to reveal rare yet meaningful patterns. In recent years, quantum computing has extended into finance, tackling tasks such as derivatives pricing, risk assessment, and portfolio optimization, all of which are critical for market forecasting [9].

Stock market instruments, spread across diverse sectors, respond differently to changes in the financial ecosystem. For instance, the energy sector suffered during the COVID-19 pandemic, while technology and consumer-based firms saw a surge in demand. The volatility in cryptocurrencies, like Bitcoin and Ethereum, further complicates financial forecasting. Traditional empirical models often struggle to predict such diverse market behavior. Quantum computing methods, such as quantum annealing (QA) and QLSTM, offer a more refined approach to solving non-convex financial optimization problems [10], potentially enhancing prediction accuracy while addressing the computational demands that challenge traditional machine learning.

A. Contributions

The following concisely describes the primary contributions of this work:

- It presents *QLSTM4FM*, a groundbreaking framework established on quantum-assisted LSTM model, which significantly enhances stock price forecasting. This framework integrates variational quantum layers that leverage quantum computing principles to optimize the learning process, allowing for more effective handling of complex temporal dependencies inherent in financial data.
- It conducts a thorough and comprehensive performance evaluation of the QLSTM model against classical LSTM models. This evaluation demonstrates the QLSTM's superiority in several critical aspects, including prediction accuracy, convergence speed, and loss reduction during both training and testing phases. The findings reveal that the QLSTM consistently outperforms traditional methods, particularly in its ability to learn intricate patterns from historical stock data more efficiently.
- Finally, it showcases the practical applicability of the QLSTM framework by implementing it to predict stock prices of Merck & Co. (MRK).

B. Organisations

The remaining sections of the manuscript are arranged as follows:

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Section II provides a review of relevant literature pertaining to our research. In Section III, we outline the proposed methodology. Section IV presents the experimental results and compares them with traditional counterparts. Finally, Section V offers concluding remarks.

II. Related Works

In contrast, quantum computers utilize qubits, which exist in superposition states of 0 and 1, providing exponential computational advantages. This allows quantum systems to process multiple inputs concurrently with fewer operations compared to classical systems [1]. Quantum algorithms have shown superior performance in areas like Fourier transforms, solving linear equations, and Eigenvalue problems [2]. Additionally, they can enhance various machine learning algorithms, including principal component analysis (PCA) [4], k-means clustering [3], support vector machines (SVMs) [5], and deep learning models like Boltzmann machines [6].

Quantum computing also simulates complex probability distributions with fewer gate operations by leveraging wave functions, a feat unattainable by classical computers. However, challenges remain, particularly in implementing nonlinearity through quantum unitaries. A promising solution is a hybrid approach that combines classical neural networks to handle nonlinearities with quantum layers for enhanced computational power. This fusion opens new possibilities for tackling complex, resource-intensive problems.

Research on financial markets has been primarily focused on predicting stock prices with accuracy, which is a goal that traders, analysts, and investors all share. Using the special properties of quantum physics, quantum computing has become an innovative instrument that can solve complex financial issues more quickly and precisely than conventional approaches. There has been a lot of interest in this novel technology, with researchers exploring its potential applications in stock price forecasting.

Traditional stock price forecasting models primarily rely on statistics-based approaches and machine learning techniques, which face limitations when dealing with large volumes of historic financial data and complex market trends. Quantum computing offers a fundamentally different framework, using concepts such as superposition and entanglement to investigate increasingly bigger solution spaces. Quantum algorithms have demonstrated the capacity to solve problems rapidly, positioning them as strong contenders for economic assessments, including stock price predictions [7]–[9].

A critical aspect of developing accurate machine learning frameworks lies in feature selection. Quantum algorithms, especially quantum annealing, excel in attribute extraction and dimensionality reduction, enhancing the precision of stock price projections and the standard of data provided. Financial modeling optimization problems can also be effectively addressed by quantum computing, with techniques like the Quantum Approximate Optimization Algorithm (QAOA) demonstrating potential in surpassing traditional equivalents [10]–[12].

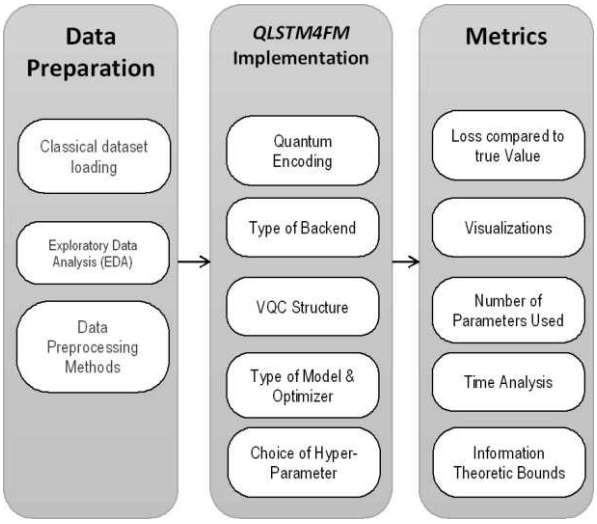

Fig. 1. *QLSTM4FM*: Proposed Framework

Quantum neural networks (QNNs) offer exponential computational acceleration and improved solution-finding capabilities [13]–[15]. Their applications span various industries, particularly in finance, where they enhance tasks like portfolio optimization and risk management.

Previous studies have examined the performance of LSTM and QLSTM models in predicting stock prices [16], [17]. Although LSTM is a good choice for identifying time-dependent trends in sequenced information, QLSTM makes use of quantum computing's benefits to increase prediction accuracy [18]–[20].

III. Proposed Methodologies

The proposed methodology for Stock Price Forecasting (SPF) employs an RNN-based approach integrated with a Variational Quantum Circuit (VQC) transformation strategy. This comprehensive framework is divided into several distinct phases as shown in Fig. 1, each designed to improve the predicted precision and efficacy of the framework. The first phase focuses on dataset creation through data collection and pre-processing. The second phase involves configuring technical indicators to capture key market signals. In the third phase, trend approximation is carried out using advanced mathematical techniques. The fourth phase compiles correlated assets to improve the model's robustness. The fifth phase applies sentiment analysis using NLP to measure market mood. Finally, the sixth phase implements and compares both QLSTM and LSTM models, leveraging quantum computing to optimize stock price predictions.

A. Dataset Creation: Data Collection and Processing

This study focuses on predicting daily closing prices for Merck & Co. Inc. (MRK) using a comprehensive dataset obtained from Yahoo Finance, spanning 2,738 trading days.

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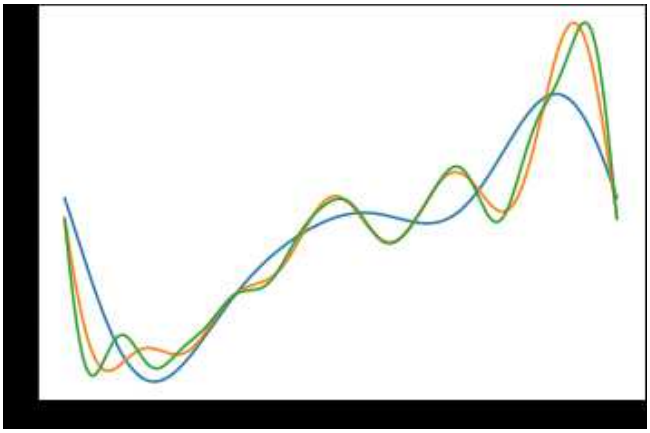

Fig. 2. Fourier transforms of MRK stock prices reveal long-term trends and short-term fluctuations based on components, 3, 6 and 9.

By leveraging both manual downloads and the Python package yfinance for real-time data extraction, we captured historical price movements alongside current market trends. The dataset includes critical metrics such as opening, closing, highest, and lowest prices, enabling us to analyze short-term fluctuations and long-term trends. Incorporating intraday price data enhances our understanding of market dynamics and volatility, which is essential for accurate predictions. Visualizations of MRK's stock price trends from 2010 to 2020 reveal significant patterns and phases, providing valuable insights for developing a robust machine learning model aimed at forecasting future price movements.

B. Configuration of Technical Indicators

In stock price forecasting, technical indicators are crucial for predicting market trends, momentum, and volatility based on historical data. Our study integrates key technical indicators as independent variables to improve model performance. These include Moving Averages (7-day and 21-day), Exponential Moving Averages (12-day and 26-day), Moving Average Convergence Divergence (MACD), Bollinger Bands, and Momentum. These indicators provide insights into stock price behavior by smoothing price fluctuations, capturing volatility, and revealing trend shifts. The analysis spans for long (2718 days) and short-term (400 days) data, helping to identify trends and trading opportunities across different time horizons. Graphical analysis supports the interpretation of crossovers, volatility shifts, and momentum changes, aiding informed investment decisions.

C. Calibrating Trend Approximation

This study utilizes Fourier Transform to analyze stock market data, specifically focusing on the decomposition of stock prices into their fundamental frequency components. By converting price data from the time domain to the frequency domain, we can filter out high-frequency noise from daily fluctuations, allowing us to retain important low-frequency components that represent long-term trends. Using Python's Fast Fourier Transform (FFT), we experiment with retaining 3, 6, and 9 frequency components to observe how this impacts our ability to approximate both long-term trends and short-term fluctuations as shown in Fig. 2. The smoothed results provide a clearer view of the stock's underlying behavior, assisting in understanding its potential movement direction. Visual comparisons of the original stock prices against the Fourier approximations demonstrate the effectiveness of this method in identifying trends. Overall, Fourier Transform proves to be a valuable analytical tool for enhancing the understanding of stock price behaviors amidst market noise.

```
dataset_total['SNP'] = list(SNP_data['Close'])
dataset_total['PFE'] = list(PFE_data['Close'])
dataset_total['JNJ'] = list(JNJ_data['Close'])
dataset_total['BMY'] = list(BMY_data['Close'])
dataset_total['VTRS'] = list(VTRS_data['Close'])
dataset_total['AMGN'] = list(AMGN_data['Close'])
dataset_total['VZ'] = list(VZ_data['Close'])
dataset_total['SNY'] = list(SNY_data['Close'])
dataset_total['GSK'] = list(GSK_data['Close'])
dataset_total['LLY'] = list(LLY_data['Close'])
dataset_total['NVS'] = list(NVS_data['Close'])
dataset_total['NYSE'] = list(NYSE_data['Close'])
dataset_total['NASDAQ'] = list(NASDAQ_data['Close'])
```

Fig. 3. Close Price forecasting of Correlated Assets

D. Compilation of Correlated Assets

To accurately predict MRK (Merck & Co., Inc.) stock movements, it is essential to consider broader pharmaceutical sector dynamics and correlated assets. We compile data on significant pharmaceutical companies as shown in Fig. 3, such as Pfizer (PFE), Johnson & Johnson (JNJ), and Bristol-Myers Squibb (BMY), as well as key market indexes like the S&P 500 and NASDAQ Composite . These correlated assets provide insights into market trends and sector-specific behaviors that can influence MRK's stock price. For instance, industry events, such as regulatory changes or major drug approvals, typically affect multiple companies simultaneously, enhancing predictive accuracy. By analyzing the performance of these correlated stocks and indexes, we can better identify trends that might not be visible when focusing solely on MRK. This comprehensive approach allows us to discern between company-specific movements and broader market sentiment, thereby improving our forecasting model's robustness and effectiveness in understanding MRK's stock behavior.

E. Sentiment Analysis through NLP

Sentiment analysis is crucial for interpreting emotions within textual data, especially in finance, where news significantly impacts stock performance. In our study, we apply FinBERT, a specialized NLP model, to analyze news articles about MRK (Merck & Co., Inc.) to gauge public sentiment and its correlation with stock price movements. Our workflow begins with data collection, extracting headlines and dates related to MRK, which we store in a CSV file. We then

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tokenize each headline using FinBERT, which categorizes sentiment into positive, negative, or neutral. By generating probabilities for each sentiment, we compile a sentiment profile for each headline. This analysis reveals insights, such as high negative sentiment probabilities indicating unfavorable news for MRK, allowing us to explore potential links between market sentiment and stock price trajectories. By integrating sentiment data with historical stock performance, we aim to understand how public perception influences MRK's stock movements.

F. QLSTM and LSTM Implementation: Training, Testing, and Forecasting

We implemented both classical Long Short-Term Memory (LSTM) and Quantum Long Short-Term Memory (QLSTM) models to predict stock prices for MRK. The classical LSTM, developed in PyTorch, establishes a baseline for performance metrics like accuracy, loss, and training speed. In the QLSTM model, quantum variational layers are integrated using Pennylane, transforming the architecture into a hybrid quantum-classical model. Classical data is converted into quantum format via amplitude and phase encoding techniques, with initial testing performed on simulators such as lightning.qubit. The workflow begins with rigorous data preparation, including loading historical stock data with PyTorch's DataLoader, performing exploratory data analysis (EDA), and normalizing variables. Both models are evaluated based on prediction accuracy, convergence speed, and computational resource usage.

This work advances our understanding of Quantum Recurrent Neural Networks (QRNNs) and shows how quantum computing might enhance conventional machine learning models for real-world uses such as financial forecasting.

The performance of both classical LSTM and QLSTM models is evaluated using key metrics such as loss function, training time, and parameter count. Mean Squared Error (MSE) is utilized as primary loss metric to calculate prediction accuracy in stock price forecasting. Additionally, training time is compared across multiple epochs to account for the computational complexity of quantum models. Parameter efficiency is analyzed, with the goal of achieving better performance using fewer parameters in QLSTM. Visualizations, including loss over epochs and predicted vs. true stock prices, will aid in understanding the models' comparative performance. Furthermore, to maximise the precision of the model and the pace of convergence, hyperparameter adjustment (e.g., learning rate, batch size) is carried out.

1) ***Classical LSTM Implementation****:* For the classical LSTM, a time-series forecasting model is implemented using PyTorch, focusing on predicting the stock prices. The stock price data is cleaned, split into train and test sets, and standardized to ascertain proper model performance. A shallow regression LSTM architecture is employed, with a sequence length of 3 days, enabling the algorithm to forecast the price for the following day. Training & testing functions handle batch processing, gradient calculation, and weight updates. The model is trained over 20 epochs, with a learning rate of 0.0001, and evaluated based on training and test losses as depicted in Experimental Results Section. Visualization of actual vs. predicted stock prices helps to assess the model's performance in real-world trading strategies.

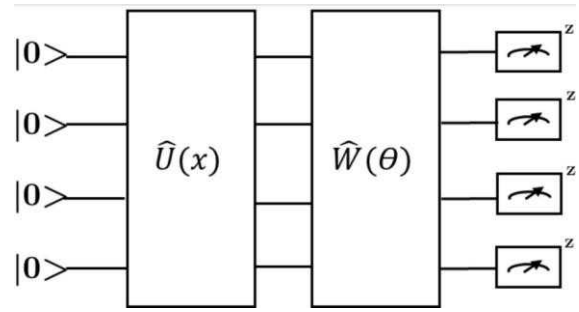

Fig. 4. QLSTM Model implemented

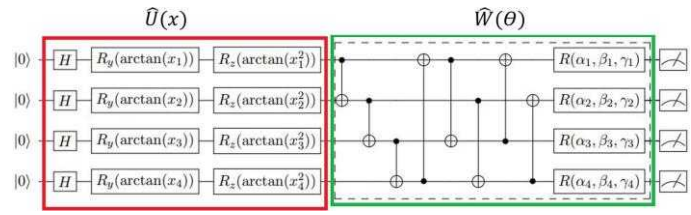

Fig. 5. Detailed visualization of Fig. 4, representing it's key components

2) ***QLSTM Implementation****:* In the QLSTM model, quantum variational layers replace key layers of the classical LSTM. This quantum-classical hybrid model is implemented using PyTorch and Pennylane. Classical data is converted to quantum format via amplitude encoding, and 4 qubits are used in the experiment as shown in Fig. 4 & 5. Training is conducted over 20 epochs, with losses tracked for both training and testing sets. The architecture involves one quantum layer with 3 rotational gates. The model's performance is visualized by plotting actual vs. predicted stock prices as depicted in Results segment, demonstrating the potential quantum advantage in improving accuracy and efficiency in stock price forecasting. Initial tests are conducted on simulators, with future plans to run on quantum hardware like IBM or AWS.

3) ***Future Work and Quantum Model Refinements****:* Future research will explore increasing the amount of qubits, running experiments on actual quantum hardware, optimizing the QLSTM architecture for better accuracy and reduced training time. Other quantum models such as Quantum GRU (QGRU) will be considered, alongside alternative quantum data encoding techniques. Theoretical studies, such as information-theoretic bounds on quantum advantage, will guide further refinement. Overall, this work shows that QLSTM can outperform classical models in stock price forecasting and opens the door for more advanced applications of quantum machine learning in financial markets.

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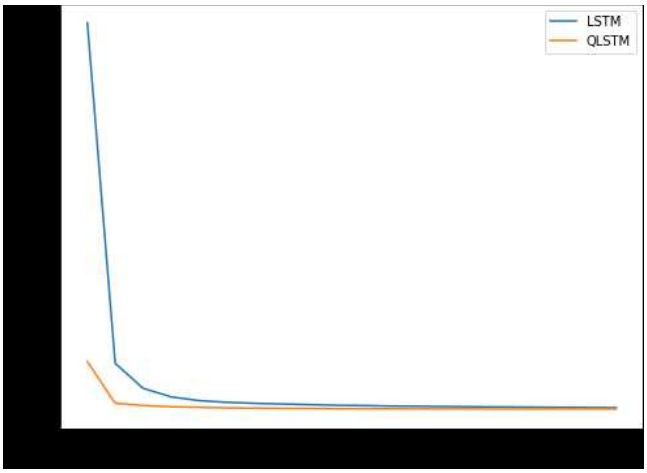

Fig. 6. LSTM vs QLSTM Training Loss

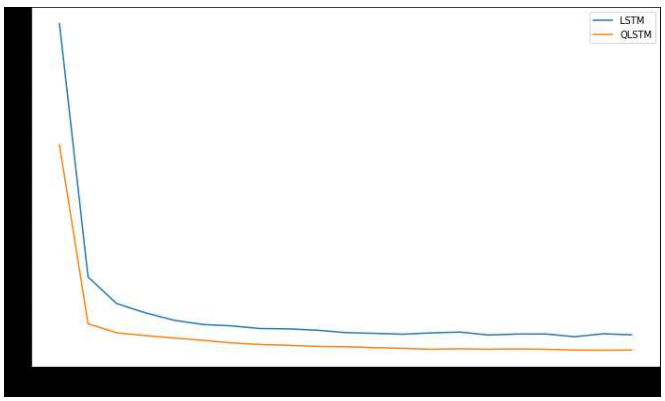

Fig. 7. LSTM vs QLSTM Testing Loss

IV. Experimental Result Analysis

A. Performance Comparison: Classical LSTM vs. QLSTM

In this section, we present a comparative analysis between the classical LSTM model and the Quantum-enhanced LSTM (QLSTM). Both models were trained on historical stock price data, and a detailed comparison reveals the QLSTM's distinct advantages in key performance metrics. Specifically, the QLSTM exhibits superior prediction accuracy, faster convergence, and enhanced parameter efficiency. While both models generate forecasts that closely align with real stock prices, QLSTM demonstrates more accurate predictions, particularly during the initial testing phases, owing to its quantum-enhanced layers that capture intricate data patterns more effectively. The QLSTM also converges more rapidly to lower loss values in both training and test phases as shown in Fig. 6 & 7, underscoring its better generalization to unseen data. Additionally, QLSTM operates with only 293 parameters, significantly fewer than the 3,217 parameters required by the classical LSTM, making it computationally efficient and suitable for real-time applications.

B. Complexity and Practicality of QLSTM

In terms of model complexity, the QLSTM is designed for implementation on Noisy Intermediate-Scale Quantum (NISQ) hardware. The framework utilized 4 qubits and 1 variational layer in this study, maintaining a linear growth in depth that ensures computational feasibility for larger datasets and future scaling. Despite the current limitations of quantum hardware, QLSTM's reduced parameter space and faster convergence make it viable for real-world applications, such as stock price prediction and other time-series forecasting tasks. The experimental results demonstrate that quantum computing, even in its nascent stages, offers a tangible advantage in machine learning tasks, delivering superior performance with fewer computational resources. This positions QLSTM as a powerful tool in predictive analytics, with potential applications in finance, healthcare, and beyond as quantum hardware continues to evolve.

C. Results

- The QLSTM model demonstrates significant advantages over classical LSTM models in prediction accuracy, convergence speed, and parameter efficiency.QLSTM consistently provides more accurate stock price predictions, especially in the initial phases of testing, maintaining closer correlations with real stock prices over time as shown in Fig. 8.
- It exhibits a faster and more substantial reduction in both training and test loss, indicating superior learning efficiency and enhanced generalisation to unobserved data. The number of parameters required for training is significantly reduced in QLSTM—only 293 compared to 3,217 for classical LSTM—minimizing.
  computational demands while preserving accuracy.
- This study highlights the potential of QLSTM for practical deployment in financial markets and various business metrics, where efficient and accurate predictions are crucial for informed decision-making.

V. Conclusion and Future Scope

This paper introduces a novel approach to stock price prediction using QLSTM networks, demonstrating that quantum machine learning can achieve comparable results to classical models with significantly fewer parameters. The findings validate QLSTM's effectiveness in financial time series analysis, highlighting its potential for efficiency in data-driven insights essential for financial markets. However, improvements are needed, such as incorporating adaptive learning rates and advanced feature selection techniques to enhance model performance. Future research could explore integrating QLSTM within Quantum Generative Adversarial Networks (QGANs) to improve predictive power and robustness. While the current model exhibited faster convergence than classical counterparts, a deeper investigation into learning rate scheduling is warranted. Overall, this study advances the application of QML in financial forecasting, underscoring the need for continued research into sophisticated quantum models and their practical

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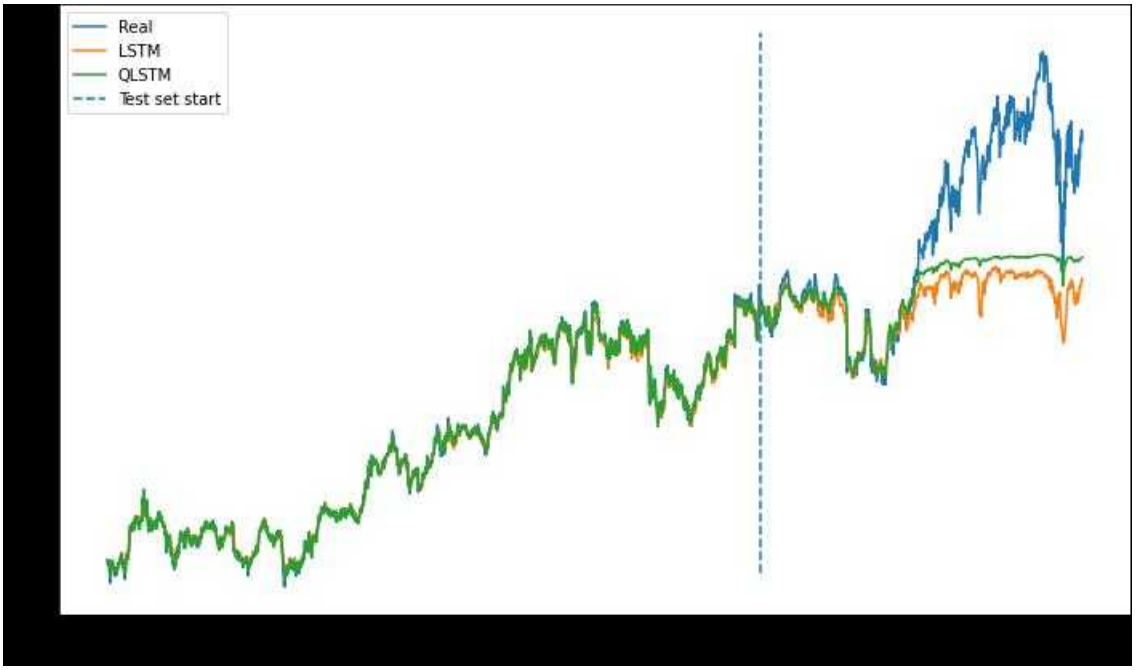

Fig. 8. Figure demonstrating QLSTM's ability to model complex stock price patterns better than LSTM over time

deployment in real-world scenarios, particularly as quantum computing technologies evolve.

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